Experiment 13.1.1: Fibonacci Sequence using Modules

1. Aim

To design and implement a Python program using modular programming to generate the Fibonacci sequence up to a given maximum value. The solution involves creating a separate module for the sequence generation logic and a main program for user interaction and validation.

2. Pseudocode

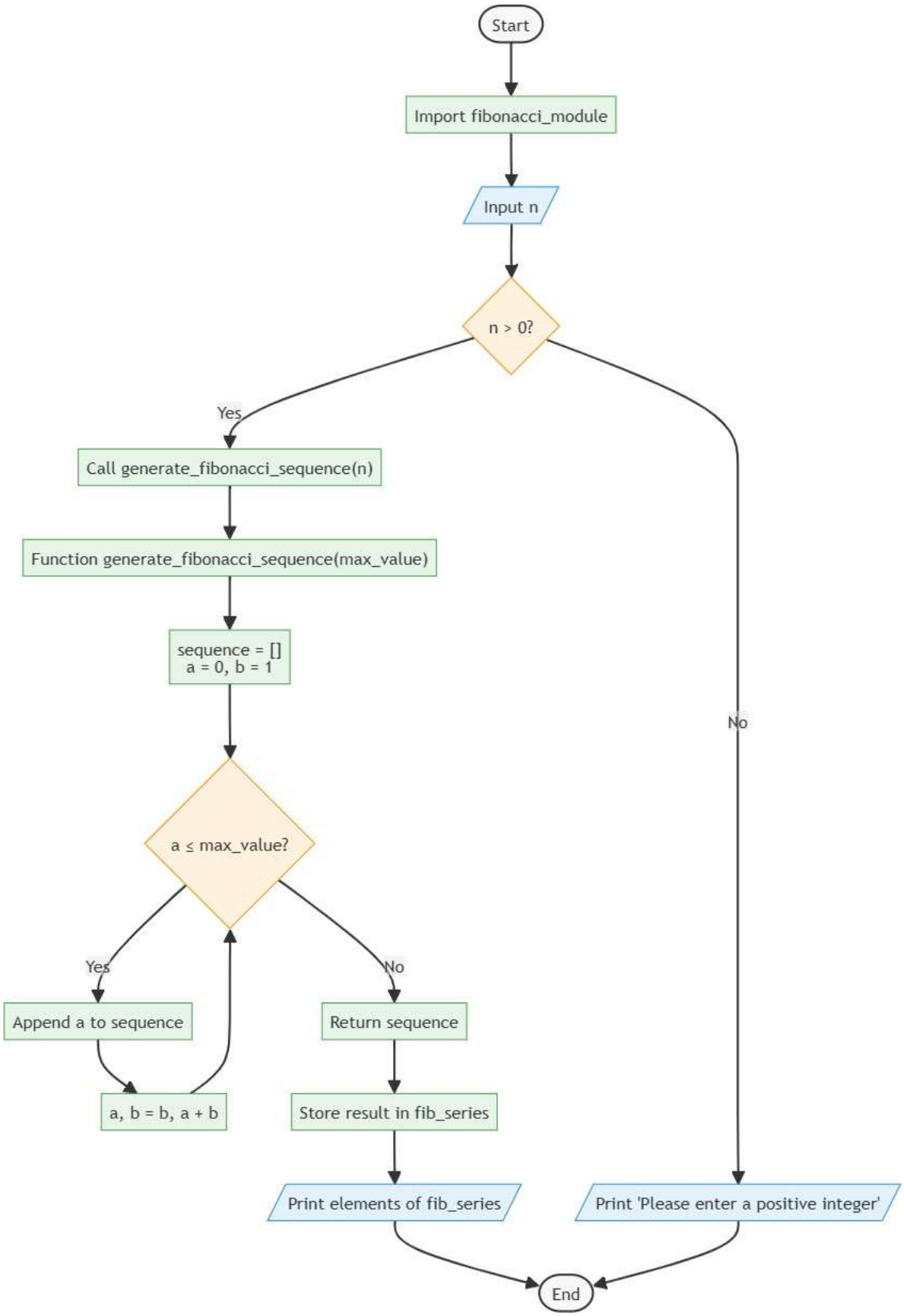
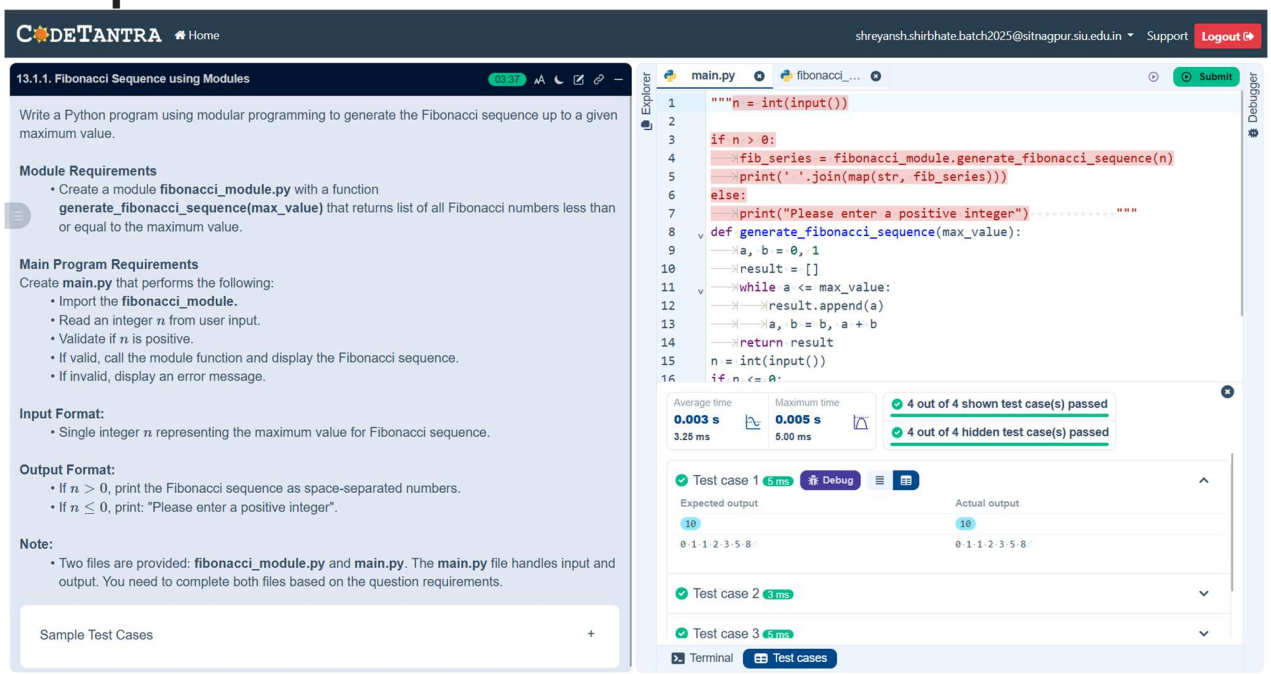
Module: fibonacci_module.py

1. **DEFINE** function generate_fibonacci_sequence(max_value):
   - **INITIALIZE** an empty list sequence.
   - **SET** $a = 0$ and $b = 1$.
   - **WHILE** $a \leq max_value$:
     - **APPEND** a to the sequence.
     - **UPDATE** a and b ($a = b, b = a + b$).
   - **RETURN** the sequence.

Main Program: main.py

1. **START**
2. **IMPORT** fibonacci_module.
3. **READ** an integer n from user input.
4. **IF** $n > 0$:
   - **CALL** fibonacci_module.generate_fibonacci_sequence(n) and store the result.
   - **PRINT** the Fibonacci sequence as space-separated numbers.
5. **ELSE**:
   - **PRINT** "Please enter a positive integer".
6. **END**

3. Flowchart


Start
Import fibonacci_module
Input n
n > 0?
Yes
No
Call generate_fibonacci_sequence(n)
Function generate_fibonacci_sequence(max_value)
sequence = []
a = 0, b = 1
a ≤ max_value?
Yes
No
Append a to sequence
a, b = b, a + b
Return sequence
Store result in fib_series
Print elements of fib_series
Print 'Please enter a positive integer'
End

4. Experiment Screenshot